Shape, Divisibility, and Stein Geometry

Strict shape, convolution rootlessness, diffusion, Legendre jets, and a corrected overlap boundary

Research report prepared for Vladimir Reshetnikov's `ProveIt` repository

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Abstract

The Fabius function, the Rvachev up-function, the Prouhet–Thue–Morse sequence, geometric infinite sinc products, and the associated q-series form a remarkably rigid dyadic system. The accompanying repository already contains an unusually large body of work on endpoint asymptotics, Lambert–W carriers, dyadic arithmetic, q-binomial acceleration, Bell–Bernoulli moment calculus, Fourier decay, representation theory, inverse-function asymptotics, Legendre and Lagrange loops, and formalization. This corrected edition separates the report's genuinely distinct shape, convolution-root, diffusion, and Legendre-jet strands from its substantial overlap with the previously filed Stein–Koopman report.

The first main result settles the critical binary endpoint of a strict log-concavity conjecture stated in the repository. The Rvachev density is already known there to be log-concave. Combining this with its nowhere-analyticity proves that it is in fact *strictly* log-concave on $(-1, 1)$. This is an unusual strictly log-concave law: every derivative of the density and of its logarithm vanishes at the mode. The theorem yields strict log-concavity of the Fabius cumulative distribution and survival functions, strict monotonicity of the hazard and reverse hazard, a strict monotone-likelihood-ratio property, and strict log-convexity of the derivative of the inverse Fabius function. The refinement equation then turns the increasing score into an explicit dyadic ratio inequality.

The second main result is a convolution-divisibility obstruction. The entire Fourier transform has simple zeros at every odd integer. Consequently the Rvachev probability law has no nontrivial identical convolution root of any order, even though it is an infinite convolution of unequal uniform laws. The same argument applies to the full geometrically scaled uniform family.

The third main strand develops the canonical Stein kernel of the centered Fabius–Rvachev distribution. It independently derives exact Fabius-tail and conditional-moment representations, exact values $\tau(0) = 5/36$ and $\tau(1/2) = 1/12$, a moment bridge to the repository's Bell–Bernoulli calculus, a weighted Poincare inequality with sharp spectral gap one, and a symmetric diffusion generator having the Rvachev law as invariant measure. The Stein kernel is nowhere analytic, but its complete Taylor jet at the mode is the quadratic $5/36 - x^2/2$. Therefore the formal eigenvalue equation of the diffusion is exactly a rescaled Legendre equation. This produces a "Legendre shadow": smooth eigenfunctions have Legendre Taylor jets, while global nonanalyticity rules out every analytic or polynomial eigenfunction beyond the constants and the linear eigenfunction.

The exact scalar-kernel formula, dyadic values, rationality and moment identities overlap *Dyadic Stein–Koopman and q-Oscillator Calculus*, already present at repository ancestor `31549ad44`; that report also proves a stronger two-term Lambert-periodic endpoint theorem. The overlap does not invalidate the proofs supplied here, but those results are not repository-distinct.

Finally, an exact logarithmic-coordinate representation and rigorous endpoint bounds are derived for the Stein kernel. They recover a Lambert–W-compatible leading equivalent already proved at greater precision in the Stein–Koopman report,

$$\tau(1-\delta) \sim \frac{\delta}{1+\delta F'(\delta)/F(\delta)} \qquad (\delta \downarrow 0),$$

and motivate a program involving multiplicative concavity, factor-divisor rigidity, Stein spectral asymptotics, strict log-concavity for the nonbinary family, and Lean formalization. A fully commented Python program and reproducible numerical certificates accompany the report.

Research-status convention

Every substantial statement is marked according to the following convention.

Imported	A fact proved in the repository or in cited literature and used here as input.
Proved here	A complete proof is supplied in this report; this label does not assert repository novelty or Lean status.
Conjectural	A precise but unproved statement, normally accompanied by structural or numerical evidence.

The submitted phrase “repository-distinct” was intended only as a corpus-relative claim, not a claim of worldwide priority. Because that source audit missed a live sibling report, this normalized edition no longer uses the phrase as a status category.

The submitted novelty audit was incomplete: it missed the existing *Dyadic Stein–Koopman and q-Oscillator Calculus* report. The corrected overlap review is pinned to repository checkpoint `563b90f25e0563cc521a53e9b478b667eaf3b694`; the earlier report is present at ancestor `31549ad44`. The labels above describe proof status inside this manuscript only. No theorem, proposition, lemma, or corollary label by itself claims an exact Lean counterpart.

The one imported input below that already has an exact Lean counterpart is the nowhere-analyticity theorem. The distinction between that formalized input and the paper-only results of this report is recorded explicitly here.

Manuscript role	Lean declaration and module	Current status
Parameterized Rvachev nowhere-analyticity input	`Fabius.rvachev_not_analyticAt` in `FabiusFunction.OriginalPaperSupplement`	Exact existing Lean theorem
Canonical Rvachev nowhere-analyticity input	`Fabius.canonical_rvachev_not_analyticAt` in `FabiusFunction.NowhereAnalytic`	Exact existing Lean theorem
Shape, rootlessness, Stein, diffusion, endpoint, and Legendre-jet results proved or proposed here	The identifiers in the suggested-name section are prospective API names only	Paper-only; no exact Lean counterpart is claimed

Contents

List of Figures

1 Orientation and principal results

1.1 Why a new direction was necessary

The repository has grown into a research library rather than a single article. Its current top-level documentation contains a primary exposition, a mathematical glossary, a non-elementarity study, a Lean walkthrough, an archive of historical sources, a paper library, and both non-formalized and semi-formalized frontier directories. The semi-formalized draft manifest records large consolidated volumes on Fourier decay, the Thue–Morse sequence, exponent sequences and q-series, Lambert W, spectra and arithmetic, integration and transforms, inverse functions and sampling, representations, exact polynomial synthesis, and recent Lagrange–Legendre–Rvachev loops [1, 2].

That breadth invalidates several initially tempting claims of novelty. In particular, the following were explicitly discarded after collision checks.

Candidate direction	Reason for exclusion from the new-results list
Non-holonomicity of the sinc product and non-P-recursiveness of moments	Already proved in the consolidated Representation Frontiers volume.
General log-concavity and the plateau transition for geometric atomic densities	Already proved in the Exponent-Sequence and q-Series Frontiers volume. Only the strict binary endpoint remained conjectural.
Fractional integration and complex-order transforms	Already treated at length in Integration and Transform Frontiers.
Fourier-shell asymptotics and log-periodic decay profiles	Covered by eight source reports and two comparative audits.
Legendre and Lagrange synthesis loops	The manifest records multiple dedicated reports, exact coefficient certificates, and consolidation work.
Bell–Bernoulli moment formulas and Lambert-series cumulants	Already part of the canonical q-series and arithmetic calculus.

The viable distinct layer is a probability-shape theory centered on *strictness*, identical-convolution rootlessness, the invariant diffusion, and its Legendre formal jets. The submitted audit's claim that the corpus had no Stein-kernel treatment was false: the Stein–Koopman report already proves the exact scalar kernel, its dyadic rational values and Appell moments, and a stronger two-term Lambert-periodic endpoint theorem. The present kernel derivations are therefore overlapping paper proofs. Strict log-concavity at the critical binary point, rootlessness, diffusion geometry, and the Legendre-jet consequences remain the report's distinct strands under the corrected checkpoint. None is promoted to Lean-proved status here.

1.2 Executive theorem map

The report proves the following chain.

R1. The Rvachev density $p = \mathrm{up}$ is strictly log-concave on $(-1, 1)$.

R2. The Fabius CDF and survival function are strictly log-concave; the hazard is strictly increasing, the reverse hazard strictly decreasing, and the location family has strict monotone likelihood ratio.

R3. The derivative of the inverse Fabius function is strictly log-convex, although its complete nonconstant Taylor jet vanishes at probability $1/2$.

R4. The score of p equals the Fabius hazard and has the exact dyadic form

$$-\frac{p'(x)}{p(x)} = \frac{2p(2x-1)}{p(x)}, \qquad 0 < x < 1.$$

It rises strictly from 0 to $+\infty$ and equals 4 at $x = 1/2$.

R5. The Rvachev law has no rth identical convolution root for any integer $r \geq 2$.

R6. The canonical Stein kernel

$$\tau(x) = \frac{1}{p(x)} \int_x^1 tp(t)\,dt$$

has exact Fabius-tail and conditional-residual-life forms, exact rational values, and Bell–Bernoulli weighted moments.

R7. The operator

$$\mathcal{L}g = \tau g'' - xg' = \frac{1}{p}(\tau p g')'$$

is symmetric in $L^2(p)$ and has exact weighted spectral gap 1.

R8. The Stein kernel is nowhere analytic, has formal center germ $5/36 - x^2/2$, and induces an exact Legendre equation on eigenfunction jets. The only analytic eigenfunctions are 1 and x.

R9. An exact endpoint integral in logarithmic coordinates gives rigorous bounds and a natural all-orders Lambert–W transseries conjecture beyond the imported two-term endpoint law for $\tau(1-\delta)$.

1.3 A compact notation table

Symbol	Meaning
U_n	independent Unif$[0,1]$ random variables
X	$\sum_{n\geq 0} 2^{-n-1}U_n$, the Fabius random variable
F	CDF of X on $[0,1]$
V_n	$2U_n - 1$, independent Unif$[-1,1]$ variables
Z	$2X - 1 = \sum_{n\geq 0} 2^{-n-1}V_n$
p	density of Z, equal to the Rvachev up-function up
H	CDF of Z
I	inverse Fabius function F^{-1}
Q	centered quantile $H^{-1} = 2I - 1$
τ	canonical Stein kernel of Z
$\rho(\delta)$	endpoint elasticity $\delta F'(\delta)/F(\delta)$
$\operatorname{sinc} z$	$\sin z/z$, with $\operatorname{sinc} 0 = 1$

2 Corpus audit and nonduplication method

2.1 Inventory strategy

The requested directory contains both active documents and historical snapshots. A literal file count by itself is not a reliable duplication audit because many old sources were later merged verbatim into canonical volumes. The audit therefore used three layers.

1. The top-level tree and all directory manifests were used to enumerate the primary exposition, archive families, paper folders, Lean material, and frontier groups.

2. Current canonical and consolidated LaTeX masters were read as the controlling mathematical state of the project. These include the primary Fabius–Rvachev exposition and the master volumes for exponents and q-series, representations, integration and transforms, inverse and sampling, spectra and arithmetic, Thue–Morse formulas, and the recent polynomial/Legendre/Lagrange reports.
3. The archive README files and provenance notes were used to map superseded LaTeX sources to the master documents into which they were absorbed. Collision searches were then run on the exact concepts needed here: strict log-concavity, Stein kernels, convolution roots, weighted Poincare inequalities, diffusion generators, and Legendre jets.
4. The correction pass also read the standalone `Fabius_Stein_Koopman_Frontier_Report` directly. This check exposed the submitted audit's missed scalar-kernel and endpoint overlap.

This method reads the corpus according to its own editorial architecture. A historical file described as a verbatim predecessor of a consolidated volume was not treated as a mathematically independent current source. Conversely, recent standalone reports were checked separately instead of being inferred absent from the consolidated masters.

2.2 Imported facts used in this report

Only a small portion of the repository's established theory is needed as input. The imported statements are:

1. $p = \mathrm{up}$ is the positive C^∞ density of Z on $(-1, 1)$, vanishes with all derivatives at ± 1, is even, and satisfies the refinement equation.
2. p is log-concave on $(-1, 1)$.
3. p is nowhere real analytic on $[-1, 1]$; in particular it is analytic on no nonempty open subinterval of $(-1, 1)$.
4. F and p take rational values at dyadic arguments, and the integration volumes provide exact dyadic antiderivative algorithms.
5. The Fourier transform has the geometric sinc product and integer zero divisor described below.
6. The centered even moments are generated by explicit Bernoulli cumulants and complete Bell polynomials.
7. The endpoint and inverse-endpoint volumes provide all-orders expansions whose scale selection is governed by Lambert W and bounded dyadic-periodic corrections.

All manuscript-level theorems below are derived from these inputs and standard named results that are cited explicitly. Their labels certify paper proofs, not novelty and not Lean formalization.

2.3 External literature boundary

The probability arguments use two classical tools. First, convolution and marginalization preserve log-concavity by the Prekopa theorem; modern surveys are [7, 8]. Second, a one-dimensional density with connected support has the canonical Stein kernel used below, and Saumard's weighted Poincare inequality applies to it [9]. These tools are general; the exact Rvachev formulas, dyadic identities, and Legendre consequences developed here do not appear in those references.

3 The dyadic probability fixed point

3.1 Random-series and refinement descriptions

Let $(U_n)_{n\geq 0}$ be independent random variables uniformly distributed on $[0,1]$, and define

$$X = \sum_{n=0}^{\infty} \frac{U_n}{2^{n+1}}. \tag{1}$$

The series converges almost surely and takes values in $[0,1]$. Its CDF is the Fabius function F. Centering and doubling gives

$$Z = 2X - 1 = \sum_{n=0}^{\infty} \frac{V_n}{2^{n+1}}, \qquad V_n = 2U_n - 1 \sim \mathrm{Unif}[-1,1]. \tag{2}$$

The density of Z is $p = \mathrm{up}$. Direct calculation gives

$$\mathbb{E}Z = 0, \qquad \mathrm{Var}(Z) = \frac{1}{9}, \qquad \mathbb{E}X = \frac{1}{2}, \qquad \mathrm{Var}(X) = \frac{1}{36}. \tag{3}$$

Splitting the first summand in (2) shows that

$$Z \overset{d}{=} \frac{V + Z'}{2}, \tag{4}$$

where $V \sim \mathrm{Unif}[-1,1]$, $Z' \overset{d}{=} Z$, and V, Z' are independent. At the density level this becomes

$$p(x) = \int_{2x-1}^{2x+1} p(t)\,dt, \qquad p(t) = 0 \quad (|t| \geq 1). \tag{5}$$

Differentiating gives

$$p'(x) = 2\big(p(2x+1) - p(2x-1)\big). \tag{6}$$

For $0 < x < 1$ this simplifies to

$$p'(x) = -2p(2x-1). \tag{7}$$

3.2 The survival identity

The affine relation $Z = 2X - 1$ and the refinement equation yield an identity that will organize much of the report.

Proposition 3.1 (Density-survival self-duality)**.** *For every $x \in [0,1]$,*

$$p(x) = \mathbb{P}(X \geq x) = 1 - F(x) = F(1-x). \tag{8}$$

Moreover,

$$F'(x) = 2p(2x-1), \qquad 0 < x < 1. \tag{9}$$

Proof. For $x \in [0,1]$, the upper endpoint in (5) lies outside the support, so

$$p(x) = \int_{2x-1}^{1} p(t)\,dt = \mathbb{P}(Z \geq 2x-1) = \mathbb{P}(X \geq x).$$

The symmetry $F(1-x) = 1 - F(x)$ gives the remaining equality. The density transformation under $Z = 2X - 1$ gives (9). □

Remark 3.2. Equation (8) says that the value of the centered density at x is exactly the survival probability of the uncentered law at the *same* number x. This accidental-looking alignment is special to the binary normalization and is the source of the hazard, residual-life, and Stein identities below.

3.3 Flatness at the mode

Because p and all its derivatives vanish at the boundary, repeated differentiation of (6) gives

$$p(0) = 1, \qquad p^{(k)}(0) = 0 \quad (k \geq 1). \tag{10}$$

Consequently every derivative of $\log p$ also vanishes at the origin. The function nevertheless decreases away from zero and is nowhere analytic. This combination will turn ordinary log-concavity into strict log-concavity without any local curvature lower bound.

4 Strict log-concavity at the critical binary point

4.1 A general strictness lemma

Lemma 4.1 (No affine patch implies strict concavity). *Let J be an interval and let $g : J \to \mathbb{R}$ be concave. If g is not affine on any nondegenerate subinterval of J, then g is strictly concave on J.*

Proof. If strict concavity fails, there are $x < y$ and $0 < t < 1$ for which

$$g((1-t)x + ty) = (1-t)g(x) + tg(y).$$

Equality at an interior point of a chord of a concave function forces equality throughout that chord, hence g is affine on $[x, y]$. □

4.2 Resolution of the binary strictness conjecture

Theorem 4.2 (Strict log-concavity of the Rvachev density). *The function $p = \mathrm{up}$ is strictly log-concave on $(-1, 1)$. Equivalently, for $x \neq y$ in $(-1, 1)$ and $0 < t < 1$,*

$$p((1-t)x + ty) > p(x)^{1-t} p(y)^{t}. \tag{11}$$

Status. PROVED HERE; this settles the $a = 2$ endpoint of the strict-log-concavity conjecture in the repository's geometric-family volume.

Proof. The repository proves that p is log-concave and positive on $(-1, 1)$, so $g = \log p$ is concave there. If g were affine on a nondegenerate open interval, then $p = e^g$ would be an exponential function on that interval and therefore real analytic there. This contradicts the imported theorem that p is nowhere analytic on $(-1, 1)$. Apply lemma 4.1. □

Corollary 4.3 (Strict but infinitely flat). *The density p is strictly log-concave, yet*

$$(\log p)^{(k)}(0) = 0 \qquad (k \geq 1). \tag{12}$$

In particular, p is not strongly log-concave on any neighborhood of the mode.

Proof. The vanishing follows from (10) and the chain rule. Strong log-concavity on a neighborhood would require $(\log p)'' \leq -c < 0$ there, which is impossible at 0. □

Remark 4.4. A common intuition identifies strict log-concavity with visibly negative second logarithmic curvature. The Rvachev density is a counterexample of maximal flatness: strictness is global and order-theoretic, while every finite-order local curvature test at the mode returns zero.

4.3 Strict log-concavity of the distribution tails

Theorem 4.5 (Strictly log-concave CDF and survival)**.** *The centered CDF $H(x) = \mathbb{P}(Z \le x)$ and survival $1 - H(x)$ are strictly log-concave on $(-1, 1)$. The Fabius CDF F and survival $1 - F$ are strictly log-concave on $(0, 1)$.*

Status. Proved here.

Proof. Prekopa's theorem implies that the integral of a log-concave density over a moving half-line is log-concave. Thus H and $1 - H$ are log-concave; affine changes of variable give the same conclusion for F and $1 - F$.

Suppose $\log H$ were affine on a nondegenerate interval. Since H is strictly increasing there, $H(x) = Ce^{ax}$ with $a > 0$, and its derivative $p(x) = aCe^{ax}$ would have affine logarithm on that interval. This contradicts theorem 4.2. The survival case is identical after differentiating Ce^{-ax}. The statements for F follow by the affine relation between X and Z. □

4.4 Hazard, reverse hazard, and likelihood ratio

Let

$$h_X(x) = \frac{F'(x)}{1 - F(x)}, \qquad r_X(x) = \frac{F'(x)}{F(x)}$$

be the hazard and reverse hazard of the Fabius law.

Theorem 4.6 (Strict reliability properties)**.** *On $(0, 1)$, the hazard h_X is strictly increasing and the reverse hazard r_X is strictly decreasing. Moreover,*

$$h_X(x) = -\frac{p'(x)}{p(x)} = \frac{2p(2x - 1)}{p(x)}. \tag{13}$$

The location family $p(x - \theta)$ has the strict monotone-likelihood-ratio property.

Status. Proved here.

Proof. Strict log-concavity of $1-F$ means $(\log(1-F))' = -h_X$ is strictly decreasing, so h_X is strictly increasing. Strict log-concavity of F similarly makes $r_X = (\log F)'$ strictly decreasing. Equation (13) follows from (8), (9), and (7).

For $\theta_2 > \theta_1$, differentiate the log likelihood ratio:

$$\frac{d}{dx} \log \frac{p(x - \theta_2)}{p(x - \theta_1)} = (\log p)'(x - \theta_2) - (\log p)'(x - \theta_1) > 0,$$

because the derivative of a differentiable strictly concave function is strictly decreasing. □

Corollary 4.7 (A strict dyadic ratio inequality)**.** *For $0 < x < y < 1$,*

$$\frac{p(2x - 1)}{p(x)} < \frac{p(2y - 1)}{p(y)}. \tag{14}$$

The common score/hazard in (13) increases from 0 to $+\infty$, and

$$-\frac{p'(1/2)}{p(1/2)} = 4. \tag{15}$$

Proof. Strict monotonicity gives (14). At 0, evenness gives $p'(0) = 0$. If the score were bounded near 1, integrating $(\log p)'$ would prevent $\log p$ from tending to $-\infty$, contradicting $p(1) = 0$. Finally, $p(0) = 1$, $p(1/2) = 1/2$, and (7) give $p'(1/2) = -2$. □

Corollary 4.8 (A refinement log-concavity inequality). *For $1/2 < x < 1$,*

$$2p(x)p(4x-3) \leq p(2x-1)^2. \tag{16}$$

Status. Proved here.

Proof. On this interval, differentiating (7) once more gives $p''(x) = 8p(4x-3)$. Pointwise log-concavity is $p(x)p''(x) - p'(x)^2 \leq 0$. Substitute $p'(x) = -2p(2x-1)$. □

4.5 The inverse Fabius function

Let $I = F^{-1} : (0,1) \to (0,1)$.

Theorem 4.9 (Strict log-convexity of the inverse slope). *The function I' is strictly log-convex on $(0,1)$. Equivalently,*

$$I'''(u)I'(u) - I''(u)^2 \geq 0, \tag{17}$$

with no nondegenerate interval of equality.

Status. Proved here; this strengthens the already known concavity/convexity split of I.

Proof. Write $f = F'$ and $\ell = \log f$. The density $f(x) = 2p(2x-1)$ is strictly log-concave. Since $I' = 1/f(I)$,

$$\frac{d^2}{du^2}\log I'(u) = \big((\ell')^2 - \ell''\big)(I(u))\, I'(u)^2 \geq 0.$$

If $\log I'$ were affine on an interval, the displayed nonnegative expression would vanish there. Hence $\ell' = 0$ and $\ell'' = 0$ throughout the corresponding I-interval, making ℓ constant there, contrary to strict log-concavity. □

Corollary 4.10 (Flat strict minimum of the inverse slope). *At $u = 1/2$,*

$$I'(1/2) = \frac{1}{2}, \qquad (I')^{(k)}(1/2) = 0 \quad (k \geq 1), \tag{18}$$

but I' is strictly log-convex and therefore cannot be constant on any interval.

Proof. The density f has value 2 at $1/2$ and all of its derivatives of positive order vanish there, because p is flat at 0. Formal inversion gives $I(1/2+h) = 1/2 + h/2+$ a flat remainder. Strict log-convexity follows from theorem 4.9. □

5 Fourier zeros and convolution power-rootlessness

5.1 The entire characteristic function

With the Fourier convention

$$\Phi(\xi) = \mathbb{E}e^{2\pi i \xi Z},$$

independence in (2) gives

$$\Phi(\xi) = \prod_{n=0}^{\infty} \operatorname{sinc}\left(\frac{\pi\xi}{2^n}\right), \qquad \operatorname{sinc} z = \frac{\sin z}{z}. \tag{19}$$

The compact support of p makes Φ entire. At a nonzero integer m, the factors indexed by $0 \le n \le \nu_2(m)$ vanish simply and all later factors are nonzero. Hence

$$\operatorname{ord}_{\xi=m} \Phi = 1 + \nu_2(m). \tag{20}$$

In particular, every odd integer is a simple zero.

Vieta's product

$$\operatorname{sinc} z = \prod_{j=1}^{\infty} \cos\left(\frac{z}{2^j}\right)$$

reorganizes (19) as

$$\Phi(\xi) = \prod_{m=1}^{\infty} \cos\left(\frac{\pi\xi}{2^m}\right)^m. \tag{21}$$

Thus Z also has the Bernoulli-triangular representation

$$Z \stackrel{d}{=} \sum_{m=1}^{\infty}\sum_{j=1}^{m} \frac{\varepsilon_{m,j}}{2^{m+1}}, \qquad \mathbb{P}(\varepsilon_{m,j} = \pm 1) = \frac{1}{2}. \tag{22}$$

This is the natural bridge from the sinc product to finite Thue–Morse sign sums.

5.2 A general zero-multiplicity obstruction

Lemma 5.1 (Entire zero divisibility). *Let μ be a compactly supported probability measure on $\mathbb{R}$. If $\mu = \nu^{*r}$ for an integer $r \ge 2$, then ν is compactly supported and every zero of the entire characteristic function $\widehat{\mu}$ has multiplicity divisible by r.*

Proof. If the support of ν were unbounded above, then for every M there would be an interval above M/r with positive ν-mass. Independence would give positive probability that all r summands lie in that interval, forcing their sum above M. This contradicts compact support of μ. The lower tail is identical, so ν has compact support.

Both characteristic functions therefore extend to entire functions, and the real identity $\widehat{\mu} = \widehat{\nu}^{\,r}$ extends to $\mathbb{C}$. At every zero, the order of the right-hand side is r times an integer. □

Theorem 5.2 (No identical convolution roots). *Let μ_{up} be the probability law with density $p = \mathrm{up}$. For every integer $r \ge 2$, there is no probability measure ν such that*

$$\mu_{\mathrm{up}} = \nu^{*r}. \tag{23}$$

Status. PROVED HERE.

Proof. The entire characteristic function Φ has a simple zero at $\xi = 1$ by (20). This contradicts lemma 5.1 for every $r \geq 2$. □

Remark 5.3 (Decomposable but power-rootless). The theorem does not say that the law is convolution-indecomposable. On the contrary, (2) writes it as an infinite convolution of unequal uniform laws, and (22) writes it as an infinite convolution of Bernoulli laws. What fails is exact divisibility into r *identically distributed* summands. This is stronger and more specific than the elementary fact that a nondegenerate compactly supported law cannot be infinitely divisible.

5.3 Extension to geometric atomic laws

Normalize the general geometric uniform series by

$$Z_a = (a-1)\sum_{n=1}^{\infty} \frac{V_n}{a^n}, \qquad a > 1, \tag{24}$$

so that $\operatorname{supp} Z_a = [-1, 1]$. Its characteristic function is a product of sinc factors with strictly decreasing scales. The first positive zero of the largest-scale factor is not a zero of any smaller-scale factor and is therefore simple.

Corollary 5.4 (Geometric-family power-rootlessness)**.** *For every $a > 1$ and every integer $r \geq 2$, the law of Z_a has no rth identical convolution root.*

Status. Proved here.

Proof. Apply lemma 5.1 at the first positive zero of the first sinc factor. □

5.4 A factor-classification conjecture

The cosine product (21) supplies many compactly supported factors: choose any submultiset of the triangular array of cosine factors, and the chosen product is the characteristic function of the corresponding Rademacher subseries. The simple-zero theorem shows that equal-factor partitions are impossible, but it leaves the general factor lattice open.

Conjecture 5.5 (Dyadic Bernoulli-divisor rigidity)**.** *Suppose $\mu_{\mathrm{up}} = \alpha * \beta$ with compactly supported probability measures α and β. Then, up to opposite translations of the two factors, there is a partition of the multiset*

$$\{(m, j) : m \geq 1,\ 1 \leq j \leq m\}$$

into $A \sqcup B$ such that

$$\widehat{\alpha}(\xi) = \prod_{(m,j)\in A} \cos\left(\frac{\pi\xi}{2^m}\right), \qquad \widehat{\beta}(\xi) = \prod_{(m,j)\in B} \cos\left(\frac{\pi\xi}{2^m}\right).$$

Status. Conjectural.

The conjecture is deliberately stated at the Bernoulli level rather than at the uniform-factor level. A uniform law itself decomposes into a Bernoulli digit and a smaller uniform tail, so any rigidity statement phrased only in terms of the original sinc factors would be false. A route toward conjecture 5.5 would combine Hadamard factorization of entire characteristic functions, positive-definiteness constraints, and the valuation-weighted zero divisor (20).

6 The canonical Rvachev Stein kernel

6.1 Definition and Stein identity

For a centered one-dimensional density p with connected support, the canonical Stein kernel is

$$\tau(x) = \frac{1}{p(x)} \int_x^1 tp(t)\,dt, \qquad -1 < x < 1. \tag{25}$$

The upper endpoint is 1 because p is supported on $[-1, 1]$.

Proposition 6.1 (Basic Stein calculus). *The kernel τ is positive, even, and C^∞ on $(-1, 1)$. It satisfies*

$$\mathbb{E}[Z\varphi(Z)] = \mathbb{E}[\tau(Z)\varphi'(Z)], \tag{26}$$

$$(p\tau)' = -xp, \tag{27}$$

$$\tau'(x) + (\log p)'(x)\tau(x) = -x. \tag{28}$$

for every smooth test function for which the expectations exist.

Status. The general kernel formula is IMPORTED; the explicit Rvachev specialization and consequences below are PROVED HERE.

Proof. Positivity is immediate because $tp(t)$ has positive integral from x to 1 for $x \geq 0$, and symmetry extends it to $x < 0$. More explicitly, mean zero and evenness give

$$\int_{-x}^1 tp(t)\,dt = \int_x^1 tp(t)\,dt,$$

so $\tau(-x) = \tau(x)$. Differentiating the numerator gives (27), and division by p yields (28). Integration by parts gives

$$\int_{-1}^1 \tau(x)\varphi'(x)p(x)\,dx = \int_{-1}^1 \varphi'(x)\left(\int_x^1 tp(t)\,dt\right)dx = \int_{-1}^1 x\varphi(x)p(x)\,dx,$$

with vanishing boundary terms. □

6.2 Exact Fabius-tail representations

The density-survival duality (8) turns the Stein kernel into a Fabius tail statistic.

Theorem 6.2 (Fabius-tail and residual-life formulas). *For $0 \leq x < 1$,*

$$\tau(x) = \frac{\displaystyle\int_0^{1-x} (1-s)F(s)\,ds}{F(1-x)}, \tag{T1}$$

$$= \frac{\displaystyle\int_x^1 t\,[1-F(t)]\,dt}{1-F(x)}, \tag{T2}$$

$$= \frac{1}{2}\mathbb{E}\big[X^2 - x^2 \mid X \geq x\big]. \tag{T3}$$

If

$$e(x) = \mathbb{E}[X - x \mid X \geq x]$$

is the mean residual life and $R_x = X - x$ under the same conditioning, then

$$\tau(x) = xe(x) + \frac{1}{2}\mathbb{E}[R_x^2], \qquad xe(x) \leq \tau(x) \leq e(x) \leq 1 - x. \tag{29}$$

Status. Proved hereindependently; the exact scalar-kernel representation also appears in the earlier Stein–Koopman report.

Proof. Substitute $s = 1 - t$ into (25) and use $p(t) = F(1-t)$ to obtain (T1); using $p(t) = 1 - F(t)$ gives (T2). Integrating by parts,

$$\int_x^1 t[1 - F(t)]\,dt = \frac{1}{2}\int_x^1 (t^2 - x^2)F'(t)\,dt.$$

Divide by $1 - F(x)$ to obtain (T3). Expanding $X^2 - x^2 = 2xR_x + R_x^2$ gives the first formula in (29). Because $0 \le R_x \le 1 - x$ and $x \le (X + x)/2 \le 1$, the remaining bounds follow. □

Remark 6.3 (A cross-scale identity). The variable x plays two roles in (T3): it is a point at which the centered density's Stein kernel is evaluated and simultaneously a threshold for the uncentered Fabius law. This is not a generic Stein-kernel formula; it is a consequence of the binary self-duality $p(x) = 1 - F(x)$.

6.3 Exact absolute moment and special values

A useful absolute moment follows directly from the distributional recursion.

Lemma 6.4 (Exact first absolute moment).

$$\mathbb{E}|Z| = \frac{5}{18}. \tag{30}$$

Proof. From (4),

$$\mathbb{E}|Z| = \frac{1}{2}\mathbb{E}|V + Z'|.$$

For fixed $z \in [-1, 1]$ and $V \sim \mathrm{Unif}[-1, 1]$,

$$\mathbb{E}|V + z| = \frac{1}{2}\int_{-1}^1 |v + z|\,dv = \frac{1 + z^2}{2}.$$

Therefore

$$\mathbb{E}|Z| = \frac{1}{4}(1 + \mathbb{E}Z^2) = \frac{1}{4}\left(1 + \frac{1}{9}\right) = \frac{5}{18}.$$

□

Theorem 6.5 (Exact Stein values). *The canonical Stein kernel satisfies*

$$\tau(0) = \frac{5}{36}, \qquad \tau\left(\frac{1}{2}\right) = \frac{1}{12}, \qquad \tau'\left(\frac{1}{2}\right) = -\frac{1}{6}, \qquad \tau''\left(\frac{1}{2}\right) = -\frac{1}{3}. \tag{31}$$

Status. Proved hereindependently; the values $\tau(0) = 5/36$ and $\tau(1/2) = 1/12$ also appear in the earlier Stein–Koopman report.

Proof. At $x = 0$, (T3) gives

$$\tau(0) = \frac{1}{2}\mathbb{E}X^2 = \frac{1}{2}\left(\operatorname{Var} X + (\mathbb{E}X)^2\right) = \frac{1}{2}\left(\frac{1}{36} + \frac{1}{4}\right) = \frac{5}{36}.$$

At $x = 1/2$, condition $X \geq 1/2$ is equivalent to $Z \geq 0$. Symmetry gives

$$\mathbb{E}[Z^2 \mid Z \geq 0] = \mathbb{E}Z^2 = \frac{1}{9}, \qquad \mathbb{E}[Z \mid Z \geq 0] = \mathbb{E}|Z| = \frac{5}{18}.$$

Using the equivalent form

$$\tau(x) = \frac{1}{8}\mathbb{E}[(1+Z)^2 \mid Z \geq 2x - 1] - \frac{x^2}{2}$$

yields $\tau(1/2) = 1/12$.

Write $s(x) = -(\log p)'(x)$. Equation (28) becomes $\tau' = s\tau - x$. At $1/2$, (15) gives $s = 4$, hence $\tau' = -1/6$. Since $p''(1/2) = 0$, one has $s'(1/2) = -(\log p)''(1/2) = 16$. Differentiating once more,

$$\tau'' = s'\tau + s\tau' - 1 = \frac{16}{12} - \frac{4}{6} - 1 = -\frac{1}{3}.$$

□

Corollary 6.6 (A rational integral identity)**.**

$$\int_0^{1/2} F(s)\,ds = \frac{5}{72}. \tag{32}$$

Proof. The tail formula for a symmetric variable gives

$$\mathbb{E}|Z| = \int_0^1 \mathbb{P}(|Z| > t)\,dt = 4\int_0^{1/2} F(s)\,ds.$$

Apply lemma 6.4. □

6.4 Dyadic rationality

The repository's exact integration machinery proves rationality of polynomially weighted Fabius integrals at dyadic endpoints. Applying that calculus to (T1) gives the following immediate manuscript corollary, independently of the same dyadic-rationality conclusion in the earlier Stein–Koopman report.

Corollary 6.7 (Dyadic rationality of the Stein kernel)**.** *If $x \in (-1, 1)$ is dyadic rational, then $\tau(x) \in \mathbb{Q}$.*

Status. Proved herefrom imported dyadic integration theorems; overlapping the earlier Stein–Koopman report.

Proof. By evenness it is enough to take $0 \leq x < 1$. The denominator $F(1-x)$ is a nonzero rational number. The numerator is the difference of the dyadic values of an antiderivative of $(1-s)F(s)$, which is rational by the repository's monomial antiderivative and dyadic evaluation results. □

6.5 Bell–Bernoulli moment bridge

Let $\mu_k = \mathbb{E}Z^k$. The repository expresses the even cumulants as

$$\kappa_{2m} = \frac{2^{2m}B_{2m}}{2m(2^{2m}-1)}, \qquad \kappa_{2m+1} = 0, \tag{33}$$

with moments recovered by complete Bell polynomials. The Stein identity adds a second layer.

Theorem 6.8 (Stein–Bell moment identity). *For every integer $m \geq 0$,*

$$\mathbb{E}[\tau(Z)Z^m] = \frac{\mu_{m+2}}{m+1}. \tag{34}$$

In particular,

$$\mathbb{E}\tau(Z) = \frac{1}{9}, \tag{35}$$

$$\mathbb{E}[\tau(Z)Z^2] = \frac{19}{2025}, \tag{36}$$

$$\mathbb{E}[\tau(Z)Z^4] = \frac{583}{297675}. \tag{37}$$

All odd mixed moments vanish.

Status. PROVED HEREindependently; the earlier Stein–Koopman report contains the corresponding Appell-moment identity.

Proof. Take $\varphi(z) = z^{m+1}/(m+1)$ in (26). Then $\varphi'(z) = z^m$, which gives (34). The imported cumulant formula (33) and the Bell-polynomial moment conversion give

$$\mu_2 = \frac{1}{9}, \qquad \mu_4 = \frac{19}{675}, \qquad \mu_6 = \frac{583}{59535}.$$

Substitution yields the displayed values. Evenness of τ and p gives the odd vanishing. □

Remark 6.9. Equation (34) turns every exact moment theorem in the existing Bell–Bernoulli calculus into an exact weighted Stein moment at no additional combinatorial cost. Conversely, numerical or analytic information on τ can be tested against a complete rational moment sequence.

7 The Rvachev–Stein diffusion and its Legendre shadow

7.1 Weighted Poincare inequality and exact gap

Saumard's one-dimensional theorem states that the canonical Stein kernel is a valid weight in a Poincare inequality for every density with connected support [9].

Theorem 7.1 (Sharp weighted Poincare inequality). *For every absolutely continuous g with finite variance,*

$$\mathrm{Var}(g(Z)) \leq \mathbb{E}\big[\tau(Z)g'(Z)^2\big]. \tag{38}$$

The optimal constant multiplying the right-hand side is exactly 1, and equality holds for every affine g.

Status. The general inequality is IMPORTED; sharpness and its Rvachev spectral interpretation are PROVED HERE.

Proof. The inequality is Saumard's weighted Poincare theorem. For $g(z) = az + b$,

$$\operatorname{Var}(g(Z)) = a^2 \operatorname{Var}(Z) = \frac{a^2}{9},$$

while (35) gives

$$\mathbb{E}[\tau(Z)g'(Z)^2] = a^2\mathbb{E}\tau(Z) = \frac{a^2}{9}.$$

Thus no smaller multiplicative constant is possible. □

7.2 A canonical symmetric generator

Define

$$\mathcal{L}g(x) = \tau(x)g''(x) - xg'(x). \tag{39}$$

Equation (27) gives the divergence form

$$\mathcal{L}g = \frac{1}{p}(\tau p g')'. \tag{40}$$

Proposition 7.2 (Symmetry, invariance, and exact first eigenvalue)**.** *For smooth f, g for which the boundary terms vanish,*

$$-\mathbb{E}[f(Z)\mathcal{L}g(Z)] = \mathbb{E}[\tau(Z)f'(Z)g'(Z)]. \tag{41}$$

Hence $\mathcal{L}$ is symmetric and nonpositive in $L^2(p)$, the Rvachev law is invariant, and

$$\mathcal{L}1 = 0, \qquad \mathcal{L}x = -x. \tag{42}$$

The nonzero weighted spectral gap is exactly 1.

Status. Proved here.

Proof. Integrate (40) by parts. The flux $\tau p = \int_x^1 tp(t)\,dt$ vanishes at both endpoints, giving (41). Taking $f = 1$ proves invariance. The identities (42) are immediate. The Rayleigh quotient associated with (41) is at least 1 by theorem 7.1, while $g(x) = x$ achieves 1. □

Formally, (39) is the generator of the diffusion

$$dY_t = -Y_t\,dt + \sqrt{2\tau(Y_t)}\,dB_t \tag{43}$$

with state space $[-1, 1]$. The coefficient degenerates at the endpoints and the invariant density vanishes there faster than any power. A complete boundary classification can be developed through the Dirichlet form rather than assumed from the formal stochastic differential equation.

7.3 The Stein kernel is nowhere analytic

Theorem 7.3 (Nonanalyticity transfer)**.** *The Stein kernel τ is real analytic on no nonempty open subinterval of* $(-1, 1)$.

Status. Proved here.

Proof. Suppose τ were analytic on an open interval $J \subset (-1, 1)$. Since $\tau > 0$ there, the Stein ODE can be rearranged as

$$\frac{p'}{p} = -\frac{x + \tau'}{\tau}. \tag{44}$$

The right-hand side is analytic on J, so

$$p(x) = C \exp\Big(-\int^x \frac{t + \tau'(t)}{\tau(t)}\, dt\Big)$$

is analytic on J. This contradicts nowhere analyticity of p. □

7.4 The exact quadratic jet at the mode

Let

$$A(x) = \int_x^1 tp(t)\, dt, \qquad \tau = A/p.$$

Since $A'(x) = -xp(x)$ and $p(x) - 1$ is flat at 0, the complete Taylor jet can be read off exactly.

Theorem 7.4 (Quadratic formal germ)**.** *At $x = 0$,*

$$\tau(0) = \frac{5}{36}, \qquad \tau'(0) = 0, \qquad \tau''(0) = -1, \qquad \tau^{(k)}(0) = 0 \quad (k \geq 3). \tag{45}$$

Equivalently,

$$\tau(x) = \frac{5}{36} - \frac{x^2}{2} + r(x), \qquad r^{(k)}(0) = 0 \quad (k \geq 0). \tag{46}$$

The remainder r is not identically zero on any neighborhood of 0.

Status. Proved here.

Proof. The identities for A give

$$A(0) = \tau(0)p(0) = \frac{5}{36}, \qquad A'(0) = 0, \qquad A''(0) = -p(0) = -1.$$

Every derivative of A of order at least 3 is a linear combination of positive order derivatives of p at 0, hence vanishes. Since $1/p - 1$ is also flat at 0, the quotient A/p has the same complete jet as A.

If r vanished on a neighborhood, then $\tau = q := 5/36 - x^2/2$ there. Substituting into $\tau p' + (\tau' + x)p = 0$ gives $qp' = 0$, hence p is constant on that neighborhood, a contradiction. □

The theorem exhibits a phenomenon parallel to the flat strict log-concavity of p: the entire formal series of τ is a quadratic polynomial, but the actual function is not analytic and is rescued from the polynomial's premature zero by a nontrivial flat correction.

7.5 Legendre equation in the eigenfunction jet

Consider a smooth solution of the formal eigenvalue problem

$$\mathcal{L}g = -\lambda g. \tag{47}$$

Substituting (46) and passing to infinite jets at 0 removes the flat remainder. The formal Taylor series of g therefore satisfies

$$\left(\frac{5}{36} - \frac{x^2}{2}\right) g'' - xg' + \lambda g = 0. \tag{48}$$

Set

$$z = \sqrt{\frac{18}{5}}\, x. \tag{49}$$

Then (48) becomes

$$(1 - z^2)y''(z) - 2zy'(z) + 2\lambda y(z) = 0, \tag{50}$$

which is Legendre's differential equation with formal degree ν determined by

$$\nu(\nu + 1) = 2\lambda. \tag{51}$$

Theorem 7.5 (Legendre-jet theorem)**.** *Let $g \in C^\infty(-1, 1)$ satisfy (47). The complete Taylor jet of g at 0, after the rescaling (49), is the jet of a solution to the Legendre equation (50). If g is even, its jet is the even local Legendre solution; if g is odd, its jet is the odd local Legendre solution.*

For the triangular values

$$\lambda_n = \frac{n(n + 1)}{2}, \tag{52}$$

the corresponding formal polynomial shadow is

$$P_n\left(\sqrt{\frac{18}{5}}\, x\right). \tag{53}$$

Status. Proved here.

Proof. Differentiate (47) arbitrarily many times at 0. Every term involving the flat remainder r in (46) vanishes. Therefore the Taylor coefficients satisfy exactly the recurrence obtained from (48). The change of variable (49) gives (50). Reflection symmetry of τ and the equation separates even and odd jets. If $2\lambda = n(n + 1)$, the standard Legendre polynomial P_n is the polynomial solution. □

Remark 7.6 (Formal, not global, Legendre structure). For $n \geq 2$, the function in (53) is not an actual eigenfunction. Its second derivative is nonzero, so the flat correction $r(x)g''(x)$ produces a nonzero, infinitely flat residual. The Legendre polynomial is the complete local jet while the true global solution, if selected by spectral boundary conditions, must differ from it by a flat function at the mode.

7.6 No higher analytic eigenfunctions

Theorem 7.7 (Analytic-eigenfunction rigidity)**.** *Suppose a nonzero solution g of (47) is real analytic on a nonempty open interval. Then either*

$$\lambda = 0 \quad \textit{and} \quad g \textit{ is constant},$$

or

$$\lambda = 1 \quad \textit{and} \quad g(x) = cx.$$

Consequently the Rvachev–Stein generator has no polynomial eigenfunctions of degree at least 2.

Status. Proved here.

Proof. If g'' is not identically zero on the analytic interval, it is nonzero on a smaller open interval. The eigenvalue equation then gives

$$\tau(x) = \frac{xg'(x) - \lambda g(x)}{g''(x)}, \tag{54}$$

which is analytic there, contradicting theorem 7.3. Hence g'' vanishes identically on the interval and $g = a + bx$ there. Substitution into (47) gives

$$-bx = -\lambda(a + bx).$$

Thus either $b = 0$ and $\lambda = 0$, or $a = 0$ and $\lambda = 1$. Interior uniqueness for a second-order linear ODE with positive leading coefficient propagates the solution throughout $(-1, 1)$. A polynomial is analytic, so the last assertion follows. □

7.7 A rational jet at the half point

At $x = 1/2$, the density has the complete formal germ

$$p\left(\frac{1}{2} + h\right) \sim \frac{1}{2} - 2h,$$

because $p(1/2) = 1/2$, $p'(1/2) = -2$, and all higher derivatives vanish there. The numerator $A = p\tau$ has

$$A\left(\frac{1}{2}\right) = \frac{1}{24}, \qquad A'(x) = -xp(x).$$

Integrating the formal product gives the following exact jet.

Proposition 7.8 (Rational half-point shadow). *The Taylor series of $\tau(1/2 + h)$ is the Taylor expansion at $h = 0$ of*

$$\frac{1 - 6h + 6h^2 + 16h^3}{12(1 - 4h)}. \tag{55}$$

In particular,

$$\tau\left(\frac{1}{2} + h\right) \sim \frac{1}{12} - \frac{h}{6} - \frac{h^2}{6} + \frac{2h^3}{3} + \frac{8h^4}{3} + \frac{32h^5}{3} + \cdots.$$

The actual Stein kernel is not equal to this rational function on any interval.

Status. Proved here.

Proof. Formally,

$$A\left(\frac{1}{2} + h\right) = \frac{1}{24} - \int_0^h \left(\frac{1}{2} + u\right)\left(\frac{1}{2} - 2u\right) du = \frac{1}{24} - \frac{h}{4} + \frac{h^2}{4} + \frac{2h^3}{3}.$$

Divide by $1/2 - 2h$. Equality on an interval would make τ analytic there, contrary to theorem 7.3. □

8 Endpoint Stein asymptotics and Lambert scales

8.1 Exact endpoint formula

Set $x = 1 - \delta$ in (T1). For $0 < \delta < 1$,

$$\tau(1-\delta) = \frac{\displaystyle\int_0^\delta (1-s)F(s)\,ds}{F(\delta)}. \tag{56}$$

Define the endpoint elasticity

$$\rho(\delta) = \delta\frac{F'(\delta)}{F(\delta)}. \tag{57}$$

For $0 < \delta < 1/2$, the Fabius equation gives the exact dyadic form

$$\rho(\delta) = 2\delta\frac{F(2\delta)}{F(\delta)}. \tag{58}$$

8.2 Tangent and chord bounds

Theorem 8.1 (Rigorous elasticity bounds)**.** *For* $0 < \delta < 1$*,*

$$\tau(1-\delta) \le \delta\frac{1-e^{-\rho(\delta)}}{\rho(\delta)} < \frac{\delta}{\rho(\delta)}. \tag{59}$$

For every $0 < \theta < 1$*, with*

$$r_\theta(\delta) = \frac{F(\theta\delta)}{F(\delta)} \in (0,1),$$

one has

$$\tau(1-\delta) \ge (1-\delta)\delta(1-\theta)\frac{1-r_\theta(\delta)}{-\log r_\theta(\delta)}. \tag{60}$$

Status. Proved here.

Proof. Let $L = \log F$, which is concave by theorem 4.5. The tangent-line bound at δ gives, for $0 < s < \delta$,

$$L(s) \le L(\delta) + L'(\delta)(s-\delta).$$

Therefore

$$\frac{1}{F(\delta)}\int_0^\delta F(s)\,ds \le \int_0^\delta e^{L'(\delta)(s-\delta)}ds = \delta\frac{1-e^{-\rho(\delta)}}{\rho(\delta)}.$$

Dropping the factor $1 - s \le 1$ in (56) proves (59).

For the lower bound, restrict the integral to $[\theta\delta, \delta]$ and use $1 - s \ge 1 - \delta$. Concavity puts L above the chord joining its endpoint values. Writing $s = \delta(\theta + (1-\theta)t)$ gives

$$\frac{F(s)}{F(\delta)} \ge r_\theta(\delta)^{1-t}.$$

Integrating $t \in [0,1]$ gives

$$\int_0^1 r^{1-t}dt = \frac{1-r}{-\log r},$$

which proves (60). $\square$

8.3 Logarithmic-coordinate representation

The endpoint structure becomes transparent after $s = \delta e^{-u}$.

Theorem 8.2 (Exact logarithmic-coordinate formula)**.** *For* $0 < \delta < 1$*,*

$$\frac{\tau(1-\delta)}{\delta} = \int_0^{\infty} (1 - \delta e^{-u}) \exp\Big(-u - \int_0^u \rho(\delta e^{-v})\, dv\Big)\, du. \tag{61}$$

Status. Proved here.

Proof. Put $s = \delta e^{-u}$ in (56). Since $ds = -\delta e^{-u} du$,

$$\frac{F(\delta e^{-u})}{F(\delta)} = \exp\big(L(\delta e^{-u}) - L(\delta)\big).$$

Differentiating $L(\delta e^{-v})$ with respect to v gives $-\rho(\delta e^{-v})$. Integration from 0 to u yields (61). □

The mass in (61) is concentrated on $u \asymp (1 + \rho(\delta))^{-1}$. If ρ varies slowly on that short logarithmic window, it can be frozen at $\rho(\delta)$. This heuristic motivated the arrival manuscript's one-term conjecture; its corrected repository status follows.

The arrival manuscript presented the next leading equivalent as a conjecture. It is not an open claim in the repository: the earlier Stein–Koopman report proves a stronger two-term Lambert-periodic endpoint theorem, and the imported endpoint elasticity identifies its lower-Lambert coordinate with $\rho(\delta)$ to leading order.

Theorem 8.3 (Endpoint Stein law, imported from the stronger two-term result)**.** *As* $\delta \downarrow 0$*,*

$$\tau(1-\delta) \sim \frac{\delta}{1 + \rho(\delta)} \sim \frac{\delta}{\rho(\delta)}. \tag{62}$$

Status. ImportedAt stronger two-term precision from the earlier Stein–Koopman report.

Conjecture 8.4 (Full endpoint transseries)**.** *Substituting the repository's all-orders Lambert–W and log-periodic expansion for* $\rho(\delta)$ *into* (61) *produces an all-orders transseries for* $\tau(1-\delta)$ *with the same dyadic phase.*

Status. Conjectural; this all-orders extension, unlike the leading and two-term endpoint laws, remains open.

8.4 Multiplicative concavity conjecture

The derivative of $t \mapsto \log F(e^t)$ is $\rho(e^t)$. Numerical evidence suggests that this logarithmic-coordinate profile is concave.

Conjecture 8.5 (Geometric log-concavity of the Fabius endpoint)**.** *The function*

$$t \longmapsto \log F(e^t), \qquad t < 0,$$

is strictly concave. Equivalently, $\rho(\delta)$ *is strictly decreasing as a function of* $\delta \in (0, 1)$*.*

Status. CONJECTURAL.

If true, $\rho(\delta e^{-v}) \geq \rho(\delta)$ in (61), immediately giving the sharper upper bound

$$\tau(1-\delta) < \frac{\delta}{1+\rho(\delta)}. \tag{63}$$

The numerical normalized ratios in table 3 approach 1 from below, exactly as this prediction requires.

9 Numerical experiments and reproducibility

9.1 Fixed-point discretization

The accompanying script `rvachev_frontier_experiments.py` computes the density directly from (5). On a uniform grid $-1 = x_0 < \cdots < x_N = 1$, start with the uniform density $p_0 = 1/2$ and iterate

$$(Tp)(x) = \int_{\max(-1,2x-1)}^{\min(1,2x+1)} p(t)\,dt. \tag{64}$$

The integral is evaluated through a cumulative trapezoidal primitive and linear interpolation. After 24 iterations on 200001 grid points, the first two derivatives are not taken by noisy finite differences; instead the exact refinement identities

$$p'(x) = 2[p(2x+1) - p(2x-1)], \tag{65}$$
$$p''(x) = 4[p'(2x+1) - p'(2x-1)] \tag{66}$$

are evaluated by interpolation. The Stein numerator is a reverse cumulative trapezoidal integral of $xp(x)$.

The script performs consistency assertions against

$$\int p = 1, \quad p(0) = 1, \quad p(1/2) = 1/2, \quad \tau(0) = 5/36, \quad \tau(1/2) = 1/12, \quad \mathbb{E}\tau = 1/9.$$

The observed absolute errors in the last three nontrivial identities are below 2×10^{-11}.

9.2 Density, score, and Stein kernel

Table 2: Selected fixed-point values. The score column is simultaneously $-p'(x)/p(x)$ and the Fabius hazard $F'(x)/(1-F(x))$.

x	$p(x)$	$\tau(x)$	score/hazard
0	1.0000000000	0.1388888889	0
0.10	0.9989165644	0.1340355216	0.0589203571
0.25	0.9305555555	0.1164562604	1.0746268657
0.50	0.5000000000	0.08333333335	4.0000000000
0.75	0.06944444447	0.03948611121	14.399999995
0.90	0.001083435626	0.01363282866	54.32396650
0.95	1.6184133×10^{-5}	0.006038886046	133.8886201
0.99	$3.5395904 \times 10^{-11}$	0.000931860552	947.5265981

The exact density values $p(1/4) = 67/72$ and $p(3/4) = 5/72$ explain the exact scores $72/67$ and $72/5$ visible in the table.

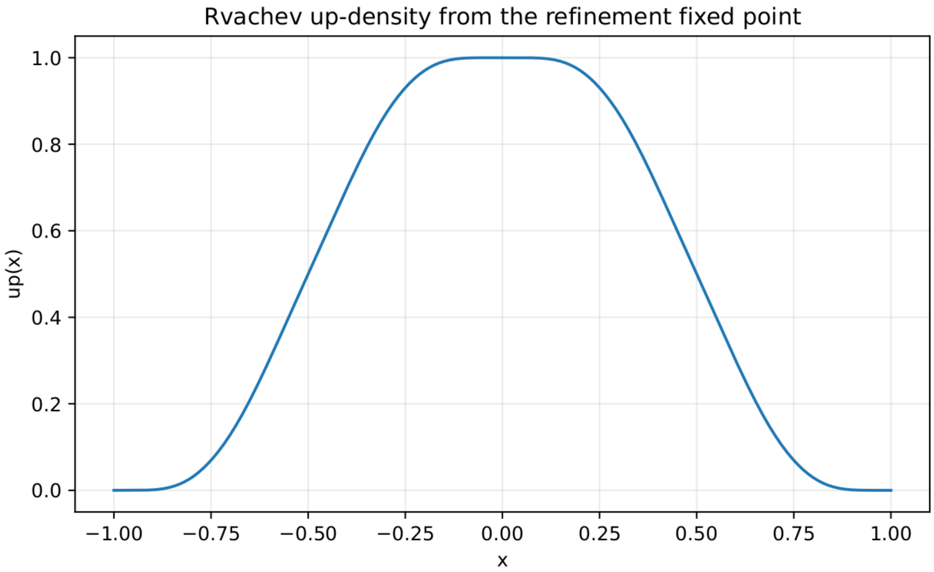

Figure 1: The Rvachev density obtained from the refinement fixed point. The graph looks rounded at the mode, although every positive-order derivative there is zero.

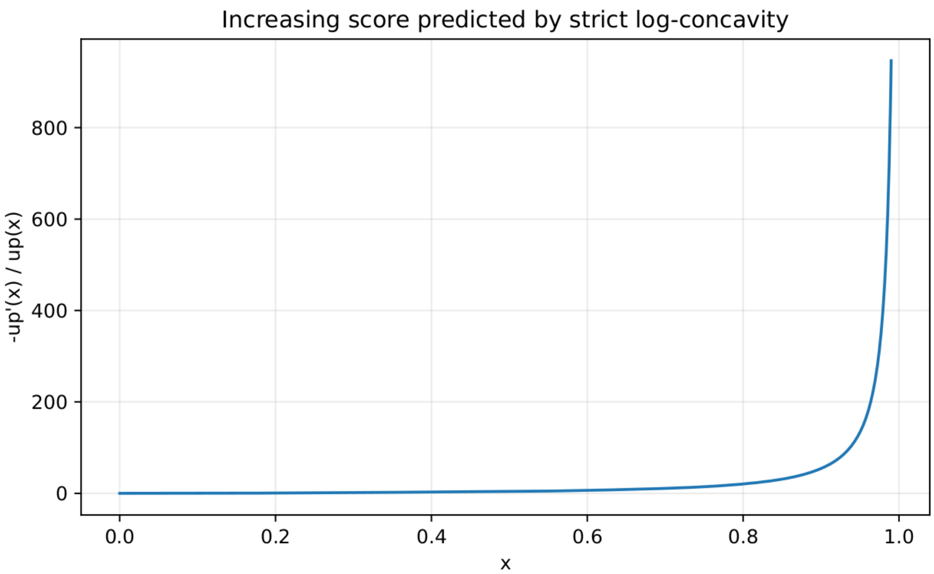

Figure 2: The score $-p'/p$, equivalently the Fabius hazard. The strict increase is proved in theorem 4.6; the plot is a diagnostic rather than evidence used in the proof.

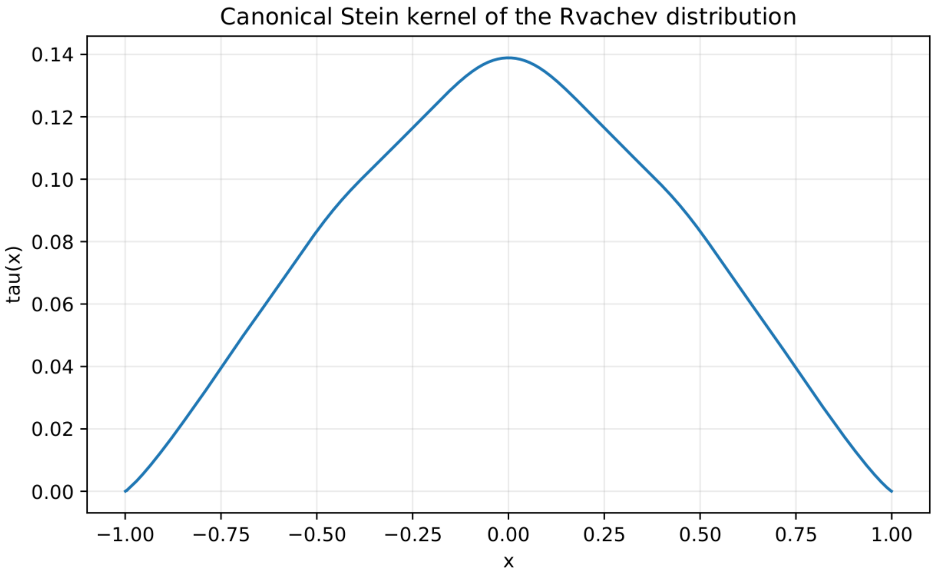

Figure 3: The even canonical Stein kernel. It has exact value $5/36$ at the mode, a quadratic infinite jet there, and a flat nonanalytic correction that keeps it positive until the endpoints.

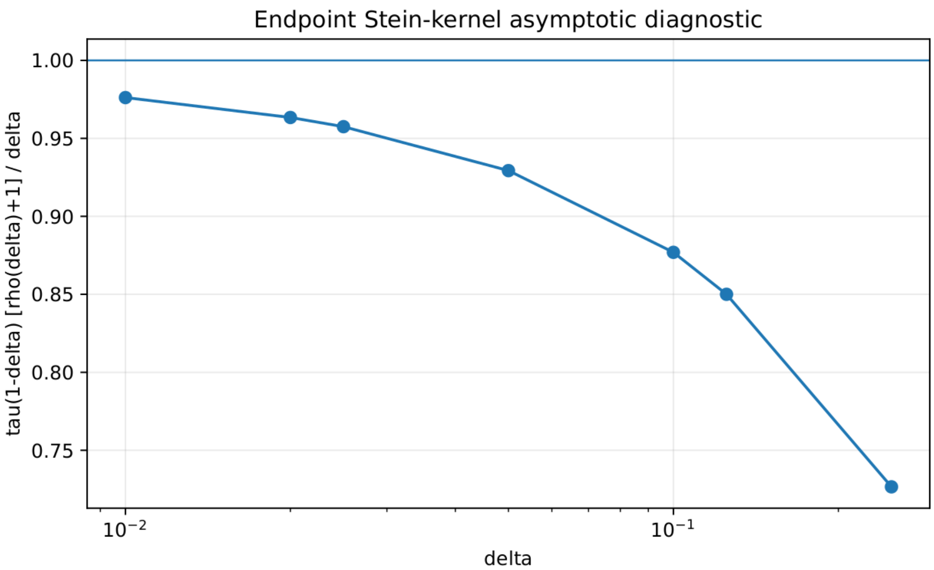

Figure 4: The ratio proved to converge to 1 by the stronger imported endpoint law summarized in theorem 8.3. The horizontal line is the theorem's limiting value.

Table 3: Endpoint Stein diagnostic. Here $\rho(\delta) = \delta F'(\delta)/F(\delta)$.

δ	$\tau(1-\delta)$	$\rho(\delta)$	$\tau(1-\delta)(\rho+1)/\delta$
0.250	0.03948611121	3.599999999	0.7265444461
0.125	0.01770833366	4.999999986	0.8500000136
0.100	0.01363282866	5.432396650	0.8769176141
0.050	0.006038886046	6.694431004	0.9293158404
0.025	0.002686175194	7.911197165	0.9574814708
0.020	0.002072559716	8.296524734	0.9633801332
0.010	0.000931860552	9.475265981	0.9761487140

9.3 Endpoint diagnostic

The second normalization $\tau(1-\delta)\rho(\delta)/\delta$ rises from about 0.569 at $\delta = 1/4$ to 0.883 at $\delta = 10^{-2}$, also consistent with the weaker equivalent in (62). Grid interpolation becomes unreliable much below $\delta = 10^{-2}$ because the density is already extremely small; endpoint asymptotic evaluation should replace brute-force gridding there.

9.4 Stein discrepancy from Gaussianity

For the centered Gaussian with the same variance, the Stein kernel is the constant $\sigma^2 = 1/9$. The computation gives

$$\mathbb{E}\tau(Z)^2 \approx 0.0129205621965, \tag{67}$$

and therefore the dimensionless quadratic Stein discrepancy

$$\frac{\mathbb{E}(\tau(Z) - 1/9)^2}{(1/9)^2} = 81\,\mathrm{Var}(\tau(Z)) \approx 0.04656553792. \tag{68}$$

This number is not claimed exact. Obtaining a closed q-series or dyadic-integral form for it is one of the concrete problems proposed below.

9.5 A monotonicity conjecture suggested by the data

The exact ODE can be written

$$\tau'(x) = h_X(x)\tau(x) - x, \qquad 0 < x < 1. \tag{69}$$

All sampled derivatives are negative, but strict decrease does not follow from ordinary log-concavity alone.

Conjecture 9.1 (Stein-kernel unimodality)**.** *The Stein kernel τ is strictly decreasing on $[0,1)$ and strictly increasing on $(-1,0]$. Equivalently,*

$$h_X(x)\tau(x) < x, \qquad 0 < x < 1. \tag{70}$$

Status. CONJECTURAL; numerically verified on the reported grid.

The kernel is not globally concave on $[0,1)$; numerical second derivatives change sign. Thus monotonicity, rather than concavity, is the plausible sharp shape statement.

10 Conjecture register and frontier program

10.1 Strictness for the full connected-support family

The geometric densities h_a from (24) are known in the repository to be log-concave for $1 < a \leq 2$, while strict log-concavity is conjectured. The proof of theorem 4.2 reduces the binary endpoint to a no-exponential- patch theorem. This suggests the following sharpened formulation.

Conjecture 10.1 (Strict geometric-family log-concavity)**.** *For every $1 < a \leq 2$, the density h_a is strictly log-concave on $(-1, 1)$. Equivalently, no h_a agrees with Ce^{bx} on a nonempty interval.*

A promising proof strategy is to insert a hypothetical exponential patch into the a-refinement equation. Differentiating propagates the patch through affine preimages. For $1 < a < 2$ those preimages overlap; one expects either a dense family of exponential patches or a finite-dimensional exponential-polynomial closure, both incompatible with compact support. This route needs less than full nowhere analyticity and may therefore work uniformly in a.

10.2 Factor divisors and Thue–Morse automata

Conjecture 5.5 asks whether every compact factor is obtained by selecting entries of the Bernoulli triangle (22). Three intermediate problems are more approachable.

Research problem (finite-level divisor classification). Classify all factorizations into characteristic polynomials of

$$\Phi_N(\xi) = \prod_{m=1}^{N} \cos\left(\frac{\pi\xi}{2^m}\right)^m.$$

Can positive definiteness be recognized by a finite Thue–Morse automaton on the zero allocations?

Research problem (uniform indecomposable atoms). Describe all compact probability factorizations of a single sinc factor after iterating Vieta's identity. Determine whether every factor is a Bernoulli digit subproduct times a translation.

Research problem (entire quotient positivity). Given an integer-valued zero divisor bounded by $1 + \nu_2(m)$, characterize when the corresponding canonical product is positive definite on $\mathbb{R}$.

A solution would connect harmonic analysis, entire functions of exponential type, and the arithmetic of the Thue–Morse cosine products.

10.3 Stein arithmetic

Corollary 6.7 gives rationality but not denominator structure. Define

$$T_{n,k} = \tau\left(\frac{k}{2^n}\right), \qquad |k| < 2^n.$$

Natural questions include:

1. exact 2-adic valuations of $T_{n,k}$;
2. common denominators across a dyadic mesh;

3. recurrences coupling $T_{n,k}$ to the repository's moment integers and Reshetnikov numbers;
4. a Bell-polynomial formula for Taylor jets of τ at arbitrary dyadic points;
5. closed forms for $\mathbb{E}\tau(Z)^m$ and in particular the discrepancy (68).

The quotient form $\tau = A/p$ indicates that denominator cancellation should be substantial. The half-point rational shadow (55) suggests that local jets may have simpler arithmetic than either numerator or denominator separately.

10.4 Stein spectral theory and Legendre comparison

The divergence-form eigenproblem is

$$-(\tau p g')' = \lambda p g, \qquad -1 < x < 1, \tag{71}$$

with natural zero-flux behavior at the endpoints. The following program would connect the new operator directly to the repository's Legendre expansions.

1. Prove closability and compactness of the Dirichlet form $\mathcal{E}(g,g) = \int \tau g'^2 p$ and obtain a simple discrete spectrum $0 = \lambda_0 < \lambda_1 = 1 < \lambda_2 < \cdots$ with alternating parity.
2. Establish a singular Sturm–Liouville Weyl law. The Liouville length is expected to be

   $$\mathscr{L} = \int_{-1}^{1} \frac{dx}{\sqrt{\tau(x)}},$$

   with $\lambda_n \sim (\pi n/\mathscr{L})^2$ if the endpoint classification permits the standard law.
3. For each eigenfunction, compute the flat correction to its Legendre jet. At the triangular formal values $n(n+1)/2$, quantify the residual $r(x)P_n''(\sqrt{18/5}\,x)$.
4. Expand the true eigenfunctions in the even/odd Legendre bases already developed in the repository and determine whether the coefficient matrix has hidden total positivity or dyadic arithmetic.
5. Compare the semigroup generated by (43) with the Ornstein–Uhlenbeck semigroup. The exact normalized Stein discrepancy is a natural first perturbative parameter.

Conjecture 10.2 (Flat Legendre correction principle)**.** *For every eigenvalue λ of the closed Rvachev–Stein operator, an eigenfunction of fixed parity equals the corresponding analytic local Legendre solution plus a nonzero function flat at 0. For $\lambda \neq 0, 1$, the flat correction is unavoidable and determines the global boundary quantization.*

10.5 Endpoint completion

The exact formula (61) reduces Stein endpoint asymptotics to control of ρ on a logarithmic window of width $O(1/\rho)$. The repository's all-orders Fabius endpoint expansions should permit the following sequence.

1. Insert the Lambert–W carrier for F into $\rho = \delta(\log F)'$ and prove a uniform slow-variation lemma.
2. Refine theorem 8.3 by proving an explicit relative error, ideally $O(1/\rho)$.
3. Transfer the bounded dyadic-periodic phase to τ and identify the first phase-dependent coefficient.

4. Feed the result into the Liouville length and endpoint classification for (71).
5. Determine whether geometric concavity conjecture 8.5 follows directly from the saddle representation.

10.6 Inverse-function consequences

Strict log-convexity of I' can be used algorithmically. On each side of $1/2$, $\log I'$ has monotone derivative, so tangent and secant bounds give certified multiplicative envelopes for the inverse slope. Combined with the existing Lambert–W endpoint seeds, this suggests globally certified Newton or Halley schemes whose step acceptance is controlled by log-convexity rather than by crude interval arithmetic.

A further analytic problem is to classify the higher logarithmic derivatives of I'. The first nontrivial inequality is (17); possible higher-order total positivity would connect the inverse Fabius function to Pick/Stieltjes or generalized Bernstein classes, though no such membership is asserted here.

11 A Lean-oriented formalization blueprint

11.1 Dependency graph

The new theory can be formalized in layers that reuse the existing Fabius Lean walkthrough.

Layer 1: imported analytic object

Formalize or import:

1. positivity and smoothness of p on $(-1, 1)$;
2. compact support, evenness, and the refinement equation;
3. log-concavity of p;
4. the theorem that p is analytic on no open subinterval.

Layer 2: strict shape

Add general library lemmas:

1. equality at one interior point of a concave chord implies affinity on the whole chord;
2. a positive function with affine logarithm is analytic;
3. strict log-concavity implies strict decrease of the logarithmic derivative;
4. Prekopa specialization for moving half-lines, or a one-dimensional proof of CDF log-concavity.

Then theorems 4.2, 4.5 and 4.6 become short compositions.

Layer 3: entire-zero obstruction

Formalize:

1. compact support of an identical convolution root;
2. entire extension of compactly supported characteristic functions;
3. multiplicativity of zero order under powers;
4. the simple zero of the first sinc factor.

The proof of theorem 5.2 then has essentially four lines.

Layer 4: Stein kernel

Define

$$A(x) = \int_x^1 tp(t)\,dt, \qquad \tau(x) = A(x)/p(x).$$

Prove differentiation, evenness, integration by parts, and the Fabius-tail change of variables. The conditional formula can be formalized without conditional expectation by first proving its numerator/denominator integral identity.

Layer 5: jets and Legendre shadow

A convenient abstraction is the ideal of functions flat at a point. Prove that it is closed under sums, products, composition with smooth functions, and division by a nonvanishing smooth germ. Then:

1. $p - 1$ is flat at 0;
2. $\tau - (5/36 - x^2/2)$ is flat;
3. the eigen-equation descends modulo the flat ideal to the Legendre equation;
4. analyticity of τ on an interval would imply analyticity of p by the first-order ODE.

This quotient-by-flat-functions viewpoint is cleaner than proving every derivative identity separately.

12 Suggested theorem names

A possible Lean namespace could include:

```
Fabius.up_strictLogConcave
Fabius.cdf_strictLogConcave
Fabius.hazard_strictMono
Fabius.inverse_deriv_strictLogConvex
Fabius.upLaw_no_convolutionPowerRoot
Fabius.steinKernel_eq_tailIntegral
Fabius.steinKernel_zero
Fabius.steinKernel_half
Fabius.stein_weightedPoincare_sharp
Fabius.steinKernel_flatGerm
Fabius.steinEigen_jet_legendre
Fabius.steinEigen_analytic_rigidity
```

13 Formalization risk ledger

Nowhere analyticity is already available in Lean as `Fabius.rvachev_not_analyticAt`, with the canonical specialization `Fabius.canonical_rvachev_not_analyticAt`; the table above records their modules. The highest remaining risk for strict log-concavity is supplying the log-concavity input and its concavity infrastructure in the exact form needed by the paper proof. The next risk is a general Prekopa theorem; for this one-dimensional setting, an elementary proof through monotone likelihood ratios may be shorter. The Stein integration identities and zero-multiplicity obstruction are comparatively low risk. None of the prospective declarations listed in the preceding section is claimed to exist in Lean.

14 Conclusions

The principal gain of this report is an organizing structure rather than one more isolated formula. The binary Fabius–Rvachev law is simultaneously:

- strictly log-concave but infinitely flat at its mode;
- a law whose density score is exactly the hazard of its own affine CDF;
- infinitely decomposable into unequal digits but power-rootless under identical convolution;
- equipped with a canonical Stein kernel having exact dyadic arithmetic and a sharp weighted spectral gap;
- governed locally by an exact Legendre jet while global nonanalyticity forbids higher analytic eigenfunctions;
- linked at the endpoint to the same Lambert–W and dyadic-periodic scales that govern the existing Fabius asymptotic theory.

These features are not independent. Nowhere analyticity upgrades log-concavity to strictness; strictness controls the hazard; the hazard enters the Stein ODE; the Stein kernel defines a diffusion; its flat jet produces the Legendre equation; and its endpoint integral transfers the Lambert scale. The resulting chain offers a compact frontier program with proof-sized intermediate goals, exact arithmetic checks, numerical diagnostics, and a realistic Lean dependency graph.

A Proof supplement: standard log-concavity facts

A.1 Convolution and weak limits

A uniform density on an interval is log-concave in the extended-value sense. Finite convolutions of log-concave functions are log-concave by Prekopa's theorem. The finite partial sums in (2) therefore have log-concave densities. Their weak limit has density p, and the pointwise/weak closure theorem for log-concave laws preserves log-concavity. This is the conceptual proof imported from the repository's geometric-family volume.

A.2 Half-line marginals

For completeness, write

$$H(x) = \int_{\mathbb{R}} \mathbf{1}_{\{t \le x\}} p(t)\, dt.$$

The set $\{(x,t) : t \le x\}$ is convex, so its indicator is log-concave in the extended-value convention. The product with $p(t)$ is log-concave in (x,t). Prekopa's theorem shows that the t-marginal $H(x)$ is log-concave. The same argument with $t \ge x$ treats the survival function.

A.3 Derivative monotonicity

If a differentiable concave function g has $g'(x) = g'(y)$ for $x < y$, monotonicity of g' forces g' to be constant on $[x,y]$, so g is affine there. Thus the derivative of a differentiable strictly concave function is strictly decreasing. This is the only strictness input needed for the score and likelihood-ratio results.

B Proof supplement: Fourier multiplicities

At an integer $m \neq 0$, the nth factor in (19) vanishes if and only if $m/2^n$ is a nonzero integer. This holds exactly for $0 \leq n \leq \nu_2(m)$. Each zero is simple because $\sin z$ has simple zeros at nonzero integer multiples of π, and the nonvanishing denominator does not change the order. The local product therefore has zero order $1 + \nu_2(m)$.

The Bernoulli triangle follows by counting pairs (n, j) with $n \geq 0$, $j \geq 1$, and $n + j = m$: there are exactly m such pairs. The support check is

$$\sum_{m=1}^{\infty} \frac{m}{2^{m+1}} = 1.$$

C Numerical file guide

The archive contains the following reproducibility files.

File	Purpose
`rvachev_frontier_experiments.py`	Fully commented fixed-point, derivative, Stein-kernel, CSV, and plotting code.
`numerical_output/pointwise_values.csv`	Values used in table 2.
`numerical_output/endpoint_diagnostics.csv`	Values used in table 3.
`numerical_output/global_diagnostics_readable.txt`	Mass, symmetry, exact-value checks, log-concavity defect, monotonicity diagnostic, and Stein discrepancy.
`numerical_output/*.png`	Deterministic raster companions embedded in this report, avoiding the arrival PDFs' Type 3 plot fonts.
`numerical_output/*.pdf`	Submitted vector figures retained as arrival evidence; they are not embedded by the normalized source.

To reproduce the outputs, run

```
python rvachev_frontier_experiments.py --output-dir numerical_output
```

from the source directory. The default computation uses NumPy and Matplotlib, writes LF-normalized tables and text, and regenerates the four PNG companions.

D Corpus map used for the audit

The most important active families encountered in the recursive documentation tree were:

1. `Fabius_Function_and_Rvachev_Up`: primary exposition and its probability, Fourier, arithmetic, asymptotic, inverse, and special-function layers;
2. `FabiusFunction_Mathematical_Glossary`;
3. `Non_Elementarity_of_the_Fabius_Function`;
4. `fabius_lean_walkthrough`;
5. `archive/`: early notes, primary exposition versions, supplements, q-calculus, standalone studies, glossary, Lean, and frontier snapshots;

6. `papers/`: Arias de Reyna, Haugland, Thue–Morse/lacunary-product, and related source papers;
7. `semi-formalized-research-frontiers/drafts/`: Fourier decay, Thue–Morse, exponents and q-series, Lambert W, spectra and arithmetic, integration and transforms, inverse and sampling, representations, polynomial synthesis, and the 2026 Lagrange/Legendre loop reports;
8. `non-formalized-research-frontiers/`: exploratory counterparts and historical frontier material.

The manifest itself states that consolidated documents absorb earlier drafts and preserve provenance hashes. The submitted audit nevertheless missed the standalone Stein–Koopman report; the corrected boundary above therefore treats its scalar-kernel and stronger endpoint results as prior corpus overlap while retaining this report's independent derivations and distinct strands.

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